Likelihood.

$$L(\pi) = \pi^k (1-\pi)^{N-k}.$$

1. **Take logs.** (Use $\log(ab) = \log a + \log b$ and $\log(x^c) = c \log x$.)

$$\begin{aligned}
\log L(\pi) &= \log\bigl(\pi^k (1-\pi)^{N-k}\bigr) \\
&= \log(\pi^k) + \log\bigl((1-\pi)^{N-k}\bigr) \\
&= k \log \pi + (N-k) \log(1-\pi).
\end{aligned}$$

2. **Differentiate** $\log L(\pi)$ **with respect to** π**.**

$$\begin{aligned}
\frac{d \log L(\pi)}{d\pi} &= k \frac{d \log \pi}{d\pi} + (N-k) \frac{d \log(1-\pi)}{d\pi} \\
&= k \left(\frac{1}{\pi}\right) + (N-k) \left(\frac{1}{1-\pi} \frac{d(1-\pi)}{d\pi}\right) \\
&= \frac{k}{\pi} - \frac{N-k}{1-\pi}.
\end{aligned}$$

3. **Set the derivative to zero at** $\pi = \hat{\pi}$ **and solve.**

$$\left.\frac{d \log L(\pi)}{d\pi}\right|_{\pi=\hat{\pi}} = \frac{k}{\hat{\pi}} - \frac{N-k}{1-\hat{\pi}} = 0.$$

$$\begin{aligned}
\frac{k}{\hat{\pi}} = \frac{N-k}{1-\hat{\pi}} &\iff k(1-\hat{\pi}) = (N-k)\hat{\pi} \\
&\iff k = N\hat{\pi} \implies \hat{\pi} = \frac{k}{N} = \frac{1}{N}\sum_{i=1}^{N} y_i.
\end{aligned}$$

4. **Second-order condition (concavity).** Need $\dfrac{d^2 \log L(\pi)}{d\pi^2} < 0.$

$$\frac{d^2 \log L(\pi)}{d\pi^2} = -\frac{k}{\pi^2} - \frac{N-k}{(1-\pi)^2} < 0 \quad (0 < \pi < 1).$$

Thus $\log L(\pi)$ is strictly concave, and the critical point $\hat{\pi} = k/N$ is the unique global maximizer.